

Solvency Field Theory V13.8

Projector-Compressed Write Operator
and Two-Phase Bridge

$\kappa + \eta_I = 1$, L_{eff} Closure,
the 3:2 Sector Accessibility Tax,
and ϕ_C/θ_g Independence

Source gates: R276O–R284O

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The same write that cannot settle its own continuation lane must grow the boundary through that lane.

One-sentence result. The microscopic write operator has a projector-compressed core with a rank-two internal closure-phase bridge: in 3+1 write capacity, one rank-one lane is reserved for continuation and three lanes are settlement-available, so the same partition gives

$$I_4 = \Pi_{\text{set}} + \Pi_{\text{cont}}, \quad \kappa = P_{\text{dir}} = \frac{3}{4}, \quad \eta_I = \frac{1}{4}, \quad \kappa + \eta_I = 1. \quad (1)$$

The leading D-scale mass coefficient is then

$$L_{\text{eff}} = \kappa(4\pi) - \frac{1}{2} \ln N_{\text{eff}} = \frac{3}{4}(4\pi) - \frac{1}{2} \ln\left(\frac{25}{13}\right) = 3\pi - \frac{1}{2} \ln\left(\frac{25}{13}\right) = 9.097814727066048 \dots \quad (2)$$

with $N_{\text{eff}} = 25/13$ from the 3:2 grouped settlement/phase participation. The V13.8 bridge: the “2” in the 3:2 split is supported by two independent closure/phase channels, $\text{rank}\{\phi_C, \theta_g\} = 2$, confirmed by independent variation of mass at fixed charge and charge at fixed mass.

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1 Claim boundary

Claimed internally.

The exterior settlement kernel is fixed by the weak-field gravity readout once energy is treated as write rate. The factor $3/4$ is derived as a projector trace ratio in 3+1 write capacity. The factor $1/4$ is the complementary continuation projector. The leading 3π in L_{eff} follows from $(3/4)(4\pi)$. The tax term $\frac{1}{2} \ln(25/13)$ follows from a 3:2 grouped sector participation: three settlement channels plus two independent closure/phase channels. The independence of ϕ_C and θ_g is supported by rank-2 variation (R283O internal, R284O public partial). The operator core is therefore:

$$\text{project} \rightarrow \text{count} \rightarrow \text{group} \rightarrow \text{tax} \rightarrow \text{compound}.$$

Still open. Full microscopic derivation of ϕ_C as native mass cadence. Full microscopic derivation of θ_g as charge/framing holonomy. Charge quantization. α and its running. Microscopic forms of K_{set} , P_{coh} , Φ_{relax} . The A_{write} raw-lane vs. readout-normalized convention.

2 Core notation

Let I_4 denote the full local 3+1 write-address capacity. Let Π_{cont} denote the rank-one continuation/tangent projector. Let Π_{set} denote the settlement-available projector:

$$I_4 = \Pi_{\text{set}} + \Pi_{\text{cont}}. \quad (3)$$

Ranks and traces:

$$\text{Tr}(I_4) = 4, \quad \text{Tr}(\Pi_{\text{cont}}) = 1, \quad \text{Tr}(\Pi_{\text{set}}) = 3. \quad (4)$$

Therefore:

$$P_{\text{dir}} = \frac{\text{Tr}(\Pi_{\text{set}})}{\text{Tr}(I_4)} = \frac{3}{4}, \quad \eta_I = \frac{\text{Tr}(\Pi_{\text{cont}})}{\text{Tr}(I_4)} = \frac{1}{4}, \quad P_{\text{dir}} + \eta_I = 1. \quad (5)$$

The historical SFT symbol κ is identified internally with this projector-class invariant: $\kappa = P_{\text{dir}} = 3/4$.

3 Count conventions

The stack uses both full-cycle and reduced/radian gauges:

Gauge	Count	Energy rate	Footprint
Full-cycle	dN_h	E/h	A_h
Reduced/radian	$dN_h = d\phi$	$E/\hbar$	$A_{\hbar}$

Consistency requires $A_h dN_h = A_{\hbar} dN_{\hbar}$, so $A_h = 2\pi A_{\hbar}$. Some existing 2π and $8\pi^2$ factors may be bookkeeping translations between gauges rather than new physics.

4 R277O: Minimal exterior settlement kernel

The exterior compactness readout is:

$$C_A(E, r) = \frac{A_{\text{write}} E}{\hbar c r}. \quad (6)$$

This is obtained when energy enters as write rate, not as extra per-write deposit. The reduced write rate is $\dot{N}_{\hbar} = E/\hbar$. The minimal retarded settlement readout:

$$C_A(x, t) = A_{\text{write}} \int \frac{\rho_E(y, t - |x-y|/c)}{\hbar c |x-y|} d^3y. \quad (7)$$

With the G-root identification $A_{\text{write}} = G\hbar/c^3$, this recovers $C_A = GM/(c^2 r)$ for rest energy $E = Mc^2$, and the gravity readout $a = -c^2 \nabla C_A$ gives $|a| = GM/r^2$.

The static exterior kernel is therefore fixed: $K_{\text{static}}(r) = 1/(cr)$.

Key bookkeeping rule: per write, one universal footprint is deposited. Per unit time, energy controls how many writes occur. Energy belongs in the write rate, not as a second factor in the per-write deposit.

5 R278O: Tangent-lane exclusion

A locally real support cannot overwrite its own continuation lane. The tangent direction is occupied by the next-write requirement:

$$\Pi_{\text{set}} = I_4 - \Pi_{\text{cont}}, \quad \text{Tr}(\Pi_{\text{set}}) = 4 - 1 = 3. \quad (8)$$

The directional settlement participation is:

$$P_{\text{dir}} = \frac{\text{Tr}(\Pi_{\text{set}})}{\text{Tr}(I_4)} = \frac{3}{4}. \quad (9)$$

Over one 4π spinor closure:

$$G_{\text{closure}} = \int_0^{4\pi} P_{\text{dir}} d\phi = \frac{3}{4}(4\pi) = 3\pi. \quad (10)$$

This replaces the older semi-geometric route $3\pi = \frac{1}{2}(6\pi)$ with the operator-native route $3\pi = \frac{3}{4}(4\pi)$.

6 R279O: $\kappa = P_{\text{dir}}$ unification

The historical $\kappa = 3/4$ is promoted to a common projector-class invariant: the same rank-one exclusion over 3+1 write capacity appears in every readout.

Readout	Full	Excluded lane	Available	Ratio
Mass compounding	4	continuation	3	$P_{\text{dir}} = 3/4$
Particle spectrum	4	occupied slot	3	$\kappa = 3/4$
SPARC morphology	4	locked lane	3	$\kappa_{\text{disk}} = 3/4$
Expansion	4	continuation	1	$\eta_I = 1/4$

The safe statement: each case realizes the same rank-one exclusion over 3+1 write capacity, not necessarily the same coordinate direction.

7 R280O: $\eta_I = 1/4$ as complement

From the same partition:

$$\eta_I = \frac{\text{Tr}(\Pi_{\text{cont}})}{\text{Tr}(I_4)} = \frac{1}{4}, \quad \kappa + \eta_I = 1. \quad (11)$$

The V13.2 quarter-boundary factor $I_{\text{write}} = \eta_I A_{\text{boundary}}/A_{\text{write}}$ is now internally derived as the continuation complement of the settlement projector.

Expansion is the continuation complement of settlement. The write that cannot settle its own continuation lane must grow the boundary through that lane.

8 R281O: Compressed operator core

The write operator is the irreversible update event. One write applies the 3+1 projector partition:

$$W : I_4 \longrightarrow \Pi_{\text{set}} \oplus \Pi_{\text{cont}}. \quad (12)$$

Settlement readout fraction: $\text{Tr}(W_{\text{set}})/\text{Tr}(W) = 3/4$.

Continuation readout fraction: $\text{Tr}(W_{\text{cont}})/\text{Tr}(W) = 1/4$.

This is not an energy split. It is a lane/readout split. One write remains one write. The partition says how that write is projected into settlement and continuation.

The schematic state update:

$$W_i[dN] : (X_i, u_i, D_{\text{set}}, C_A, I_{\text{write}}) \mapsto (X'_i, u'_i, D'_{\text{set}}, C'_A, I'_{\text{write}}). \quad (13)$$

1. **Admissibility:** $X'_i = \text{next legal no-overwrite address}$.
2. **One-speed:** support advancement respects the c -budget.
3. **Settlement projection:** $D'_{\text{set}} = D_{\text{set}} + \Pi_{\text{set}} K_{\text{set}} dN - \Phi_{\text{relax}}$.
4. **Continuation projection:** $I'_{\text{write}} = I_{\text{write}} + \Pi_{\text{cont}} dN$.
5. **Gravity readout:** $a = -c^2 \nabla C_A$.
6. **Mass/closure readout:** $\Delta \ln D_{\text{set}} = P_{\text{coh}} L_{\text{eff}} u - \Phi_{\text{relax}}$.

The compressed operator core:

$$\boxed{W = \text{irreversible write} + (\Pi_{\text{set}}, \Pi_{\text{cont}}) + \text{no-overwrite admissibility}.} \quad (14)$$

9 R282O: The 3:2 sector tax and 25/13

The settlement projector has $\text{rank}(\Pi_{\text{set}}) = 3$. These are the three spatial settlement channels. The closure supplies two independent internal phase channels:

$$\phi_C = \text{mass cadence} / \text{Compton closure phase}, \quad (15)$$

$$\theta_g = \text{charge/framing holonomy phase}. \quad (16)$$

The participation structure groups as 3:2 — three settlement components and two phase components, not five spacetime dimensions. The grouped weights:

$$w_S = \frac{3}{5}, \quad w_\Phi = \frac{2}{5}. \quad (17)$$

The participation concentration:

$$\chi = w_S^2 + w_\Phi^2 = \frac{9}{25} + \frac{4}{25} = \frac{13}{25}. \quad (18)$$

The effective participation number:

$$N_{\text{eff}} = \frac{1}{\chi} = \frac{25}{13}. \quad (19)$$

The amplitude-level accessibility tax:

$$T_{\text{amp}} = \frac{1}{2} \ln N_{\text{eff}} = \frac{1}{2} \ln \left(\frac{25}{13} \right) = 0.326963233703332 \dots \quad (20)$$

Uniqueness check. If settlement rank is 3 and internal phase count is n , then $N_{\text{eff}}(n) = (3+n)^2/(9+n^2)$. Requiring $N_{\text{eff}} = 25/13$ selects $n = 2$ as the unique integer solution, matching ϕ_C and θ_g .

10 R283O: ϕ_C/θ_g independence (internal)

The 3:2 sector tax requires two independent internal phase channels. R283O tests whether ϕ_C (mass cadence) and θ_g (charge holonomy) are genuinely independent or could be aspects of one hidden phase.

Two independent physical variations are available:

Same charge, different mass ($\delta\theta_g = 0$, $\delta\phi_C \neq 0$): the electron, muon, and tau all carry charge $-e$ but have masses differing by factors of 207 and 3477. θ_g is held fixed while ϕ_C varies freely.

Same mass, opposite charge ($\delta\phi_C = 0$, $\delta\theta_g \neq 0$): the electron and positron have identical mass but opposite charge. ϕ_C is held fixed while θ_g reverses.

Therefore:

$$\text{rank}\{\phi_C, \theta_g\} = 2. \quad (21)$$

If the two phases were the same variable, changing one would necessarily change the other. The physical particle inventory rules this out.

11 R284O: Public independence bridge

The public-facing version of the independence argument supports the 3:2 sector split directly:

$$3:2 = \text{rank}(\Pi_{\text{set}}) : \text{rank}\{\phi_C, \theta_g\} = 3 : 2. \quad (22)$$

R278O–R281O derive $\text{rank}(\Pi_{\text{set}}) = 3$. R283O/R284O support $\text{rank}\{\phi_C, \theta_g\} = 2$.

Therefore the sector participation space is 3:2, giving $N_{\text{eff}} = 25/13$ and $T_{\text{amp}} = \frac{1}{2} \ln(25/13)$.

Public status: independence shown; microscopic phase origins still open.

12 Full L_{eff} derivation

From the operator-native form:

$$L_{\text{eff}} = \kappa(4\pi) - T_{\text{amp}}. \quad (23)$$

Insert the projector result $\kappa = 3/4$:

$$\kappa(4\pi) = \frac{3}{4}(4\pi) = 3\pi = 9.42477796076938\dots \quad (24)$$

Insert the sector tax:

$$T_{\text{amp}} = \frac{1}{2} \ln\left(\frac{25}{13}\right) = 0.326963233703332\dots \quad (25)$$

Therefore:

$$L_{\text{eff}} = 3\pi - \frac{1}{2} \ln\left(\frac{25}{13}\right) = 9.097814727066048\dots \quad (26)$$

Every coefficient traced to geometry:

Piece	Value	Origin
3/4	$\kappa = P_{\text{dir}}$	Tangent-lane exclusion in 3+1
4π	spinor closure	4π closure geometry (V13.0)
25/13	N_{eff}	3:2 settlement/phase grouping
$\frac{1}{2} \ln$	amplitude tax	Grouped participation penalty

Zero free parameters. Every coefficient is dimensional counting or closure geometry.

13 Interpretation by readout

Mass compounding. $\Delta \ln D_{\text{set}} = P_{\text{coh}} L_{\text{eff}} u - \Phi_{\text{relax}}$. The leading gain is settlement-available spinor closure phase. The tax is the amplitude-level penalty from the 3:2 grouped participation.

Gravity. $a = -c^2 \nabla C_A$ with $C_A = A_{\text{write}} E / (\hbar c r)$ and $A_{\text{write}} = G \hbar / c^3$. The Newtonian exterior limit follows from the static kernel.

Expansion. $H = \dot{N}_{\text{write}} / I_{\text{write}}$ with $I_{\text{write}} = \eta_I A_{\text{boundary}} / A_{\text{write}}$ and $\eta_I = 1/4$. Expansion is the continuation complement of settlement.

Particle spectrum and galactic morphology. The historical $\kappa = 3/4$ appearances are internally reinterpreted as the same projector-class invariant: $\kappa = \text{Tr}(\Pi_{\text{available}}) / \text{Tr}(I_4) = 3/4$.

14 Open debts

The projector partition, L_{eff} closure, and two-phase independence bridge are earned. The subspace dynamics within the projected lanes remain open:

K_{set} — exact microscopic deposition kernel beyond the exterior static limit.

P_{coh} — microscopic coherence projector beyond the galactic proxy p_{gas} .

Φ_{relax} — local relaxation/dilution law under expansion without erasing history.

A_{write} — raw-lane vs. readout-normalized convention for public presentation.

ϕ_C microscopic origin — full derivation of mass cadence from closed support phase.

θ_g microscopic origin — full derivation of charge/framing holonomy from directional write orientation.

Charge quantization and fractional quark winding. α running and θ_g gauge dynamics.

Final summary

R276O–R284O convert the settlement-operator problem from a broad unknown functional into a compact projector-and-phase problem. The result is:

$$\boxed{\text{The write operator is a 3+1 projector partition applied to irreversible writes,}} \quad (27)$$

$$\boxed{\text{with a rank-two internal closure-phase bridge.}} \quad (28)$$

Three lanes settle. One lane continues. The settlement lanes generate mass, gravity, and structure. The continuation lane generates boundary growth and expansion. The 4π spinor closure supplies the phase length. The 3:2 settlement/phase grouping supplies the accessibility tax. The rank-two phase bridge supplies the two in that 3:2 split. Together they give the frozen L_{eff} with no adjustable number.

The V13.8 addition:

The three settlement lanes and two independent closure phases form the protected 3:2 participation space that taxes the spinor closure gain into the observed L_{eff} .

Things look like the geometry that created them.

Acknowledgments

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